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VEHICLE HAVING ADJUSTABLE SUSPENSION

Abstract

A damping control system for a vehicle having a suspension located between a plurality of ground engaging members and a vehicle frame includes at least one adjustable shock absorber having an adjustable damping profile.

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Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/392,063, filed Dec. 21, 2023, which application is a continuation of U.S. patent application Ser. No. 17/355,650, filed Jun. 23, 2021, issued as U.S. Pat. No. 11,878,678, which application is a divisional of U.S. patent application Ser. No. 15/816,368, filed Nov. 17, 2017, issued as U.S. Pat. No. 11,110,913, which claims the benefit of priority to U.S. Provisional Application Ser. No. 62/424,285, filed Nov. 18, 2016 and U.S. Provisional Application Ser. No. 62/524,192, filed Jun. 23, 2017, the entire disclosures of which are expressly incorporated by reference herein.

BACKGROUND AND SUMMARY OF THE DISCLOSURE

[0002] The present disclosure relates to improved suspension for a vehicle and in particular to systems and methods of damping control for shock absorbers.

[0003] Currently some off-road vehicles include adjustable shock absorbers. These adjustments include spring preload, high and low speed compression damping and/or rebound damping. In order to make these adjustments, the vehicle is stopped and the operator makes an adjustment at each shock absorber location on the vehicle. A tool is often required for the adjustment. Some on-road automobiles also include adjustable electric shocks along with sensors for active ride control systems. The system of the present disclosure allows an operator to make real time “on-the-go” adjustments to the shocks to obtain the most comfortable ride for given terrain and payload scenarios.

[0004] Exemplary systems are disclosed in U.S. Pat. No. 9,010,768 and US Published Patent Application No. 2016/0059660, both assigned to the present assignee and the entire disclosures of each expressly incorporated by reference herein.

[0005] Vehicles often have springs (coil, leaf, or air) at each wheel, track, or ski to support a majority of the load. The vehicle of the present disclosure also has electronic shocks controlling the dynamic movement of each wheel, ski, or track. The electronic shocks have one or more valves that control the damping force of each shock. This valve may control compression damping only, rebound damping only, or a combination of compression and rebound damping. The valve(s) may be connected to a controller having a user interface that is within the driver's reach for adjustment while operating the vehicle. In one embodiment, the controller increases or decreases the damping characteristics of the shock absorbers based on user inputs received from an operator. In another embodiment, the controller has several preset damping modes for selection by the operator. The controller is also coupled to sensors supported by at least one of the suspension and the chassis to provide an actively controlled damping system.

[0006] In an exemplary embodiment of the present disclosure, a recreational vehicle is provided. The recreational vehicle comprising a plurality of ground engaging members; a frame supported by the plurality of ground engaging members; at least one suspension coupling the plurality of ground engaging members to the frame, the at least one suspension including at least one adjustable shock absorber having at least one adjustable damping profile; a driveline torque management system operatively coupled to at least one of the plurality of ground engaging members, the driveline torque management system having at least one adjustable driveline torque management system characteristic; at least one airborne sensor positioned on the recreational vehicle, the at least one airborne sensor monitoring a movement characteristic of the recreational vehicle; and at least one controller operatively coupled to the driveline torque management system, the at least one airborne sensor, and the at least one adjustable shock absorber. The at least one controller determining if the recreational vehicle is in an airborne state and in response to determining the recreational vehicle is in the airborne state altering both the adjustable damping profile of the at least one adjustable shock absorber and altering the at least one adjustable driveline torque management system of the driveline torque management system.

[0007] In another exemplary embodiment of the present disclosure, a recreational vehicle is provided. The recreational vehicle comprising a plurality of ground engaging members; a frame supported by the plurality of ground engaging members; at least one suspension coupling the plurality of ground engaging members to the frame, the at least one suspension including at least one adjustable shock absorber configurable in a plurality of ride modes each having an associated damping profile; a plurality of vehicle condition sensors supported by the plurality of ground engaging members; and at least one controller operatively coupled to the at least one adjustable shock absorber and the plurality of vehicle condition sensors. The at least one controller receives a plurality of inputs from the plurality of vehicle condition sensors. The at least one controller determines if the at least one adjustable shock may be reconfigured from a first ride mode of the plurality of ride modes to a second ride mode of the plurality of ride modes by determining if a first condition is satisfied, if the first condition is satisfied then configuring the at least one shock absorber in the second ride mode, if the first condition is not satisfied then determining if a second condition is satisfied, if the second condition is satisfied then configuring the at least one shock absorber in the second ride mode, if both the first condition and the second condition are not satisfied remaining in the first ride mode, wherein only one of the first condition and the second condition is based on at least one of a plurality of inputs from the plurality of vehicle condition sensors.

[0008] In still another exemplary embodiment of the present disclosure, a recreational vehicle is provided. The recreational vehicle comprising a plurality of ground engaging members; a frame supported by the plurality of ground engaging members; at least one suspension coupling the plurality of ground engaging members to the frame, the at least one suspension including at least one adjustable shock absorber having at least one adjustable damping profile; a driveline torque management system operatively coupled to at least one of the plurality of ground engaging

members, the driveline torque management system having at least one adjustable driveline torque management system characteristic; at least one airborne sensor positioned on the recreational vehicle, the at least one airborne sensor monitoring a movement characteristic of the recreational vehicle; at least one controller operatively coupled to the driveline torque management system, the at least one airborne sensor, and the at least one adjustable shock absorber. The at least one controller determining if the recreational vehicle is in an airborne state and in response to determining the recreational vehicle is in the airborne state altering the adjustable damping profile of the at least one adjustable shock absorber to a first damping profile if the airborne state spans a first time duration and to a second damping profile if the airborne state spans a second time interval, the second time interval being longer than the first time interval.

[0009] In a further exemplary embodiment of the present disclosure, a method of controlling a damping characteristic of at least one adjustable shock absorber of a vehicle being operated by a driver is provided. The method comprising determining the vehicle is one of cornering and spinning based on a yaw characteristic of the vehicle; and adjusting the damping characteristic of the at least one adjustable shock absorber of the vehicle based on the yaw characteristic.

[0010] In yet a further exemplary embodiment of the present disclosure, a method of controlling a damping characteristic of at least one adjustable shock absorber of a vehicle being operated by a driver is provided. The method comprising (a) receiving a plurality of inputs from a plurality of sensors supported by the vehicle, the plurality of sensors including a first sensor having a first input and a second sensor having a second input; (b) based on at least the first input from the first sensor, reactively determining if the vehicle is cornering; (c) based on at least the second input from the second sensor, predictively determining if the vehicle is cornering; and (d) adjusting the damping characteristic of the at least one adjustable shock absorber of the vehicle when it is determined that the vehicle is cornering in either step (b) or step (c).

[0011] In still another exemplary embodiment of the present disclosure, a recreational vehicle is provided. The recreational vehicle comprising a plurality of ground engaging members; a frame supported by the plurality of ground engaging members; at least one suspension coupling the plurality of ground engaging members to the frame, the at least one suspension including at least one adjustable shock absorber configurable in a plurality of ride modes each having an associated damping profile; a plurality of vehicle condition sensors supported by the plurality of ground engaging members, the plurality of vehicle condition sensors including a yaw characteristic sensor; and at least one controller operatively coupled to the at least one adjustable shock absorber and the plurality of vehicle condition sensors. The at least one controller receiving a plurality of inputs from the plurality of vehicle condition sensors and configured to determine when the vehicle is cornering based on an input from the yaw characteristic sensor.

[0012] In still yet a further exemplary embodiment of the present disclosure, a recreational vehicle is provided. The recreational vehicle comprising a plurality of ground engaging members; a frame supported by the plurality of ground engaging members; at least one suspension coupling the plurality of ground engaging members to the frame, the at least one suspension including at least one adjustable shock absorber configurable in a plurality of ride modes each having an associated damping profile; a plurality of vehicle condition sensors supported by the plurality of ground engaging members, the plurality of vehicle condition sensors including an ambient weather sensor; and at least one controller operatively coupled to the at least one adjustable shock absorber and the plurality of vehicle condition sensors. The at least one controller receiving a plurality of inputs from the plurality of vehicle condition sensors and adjusting at least one damping characteristic of the at least one adjustable shock based on an input from the ambient weather sensor.

[0013] In yet another exemplary embodiment of the present disclosure, a method of controlling a damping characteristic of at least one adjustable shock absorber of a vehicle being operated by a driver is provided. The method comprising receiving an indication of an ambient weather condition of an environment surrounding the vehicle; and adjusting the damping characteristic of the at least

one adjustable shock absorber of the vehicle based on the indication of the received ambient weather condition.

[0014] In yet still another exemplary embodiment of the present disclosure, a method of controlling a damping characteristic of at least one adjustable shock absorber of a vehicle being operated by a driver is provided. The method comprising sensing a change in a terrain being traversed by the vehicle; and adjusting the damping characteristic of the at least one adjustable shock absorber of the vehicle based on the sensed change in the terrain.

[0015] In still another exemplary embodiment of the present disclosure, a method of controlling a damping characteristic of at least one adjustable shock absorber of a vehicle being operated by a driver is provided. The method comprising storing at least one historical suspension damping characteristic and at least one vehicle location corresponding to the at least one historical suspension damping characteristic; and adjusting the damping characteristic of the at least one adjustable shock absorber at a first vehicle location based on the at least one historical suspension damping characteristic corresponding to the first vehicle location.

[0016] In yet another exemplary embodiment of the present disclosure, a recreational vehicle for operation by an operator is provided. The recreational vehicle comprising a plurality of ground engaging members; a frame supported by the plurality of ground engaging members; at least one suspension coupling the plurality of ground engaging members to the frame, the at least one suspension including at least one adjustable shock absorber having at least one adjustable damping profile; a plurality of vehicle condition sensors supported by the plurality of ground engaging members; at least one controller operatively coupled to the at least one adjustable shock absorber and the plurality of vehicle condition sensors, the at least one controller receiving a plurality of inputs from the plurality of vehicle condition sensors; and a user interface system supported by the frame. The user interface system including a display. The display being configurable with the at least one controller to display a screen layout, the screen layout including at least one of (i) at least one numerical indication of at least one damping characteristic of the at least one adjustable shock absorber and (ii) at least one graphical representation of the at least one damping characteristic of the at least one adjustable shock absorber, the screen layout further including at least one of (i) a notification of an active vehicle condition modifier which alters the at least one damping characteristic of the at least one adjustable shock absorber and (ii) a vehicle steering angle indicator.

[0017] In an example thereof, the at least one damping characteristic relates to a compression damping value of the at least one adjustable shock.

[0018] In another example thereof, the at least one damping characteristic relates to a rebound damping value of the at least one adjustable shock.

[0019] In yet another example thereof, the screen layout further includes a vehicle g-force indicator, the vehicle g-force indicator provides a first indication of a magnitude of the g-force being experienced by the vehicle and a second indication of a direction of the g-force being experienced by the vehicle.

[0020] In still another example thereof, the screen layout further a graphical representation of the vehicle, the at least one adjustable shock absorber includes a first shock absorber associated with a first suspension of the at least one suspension and a second shock absorber associated with a second suspension of the at least one suspension, the first shock absorber being positioned on a first side of a vertical longitudinal vehicle centerline plane and the second shock absorber being positioned on a second side of the vertical longitudinal vehicle centerline plane, the second side being opposite the first side. In a variation thereof, a first numerical indication of the at least one numerical indication is associated with the first shock absorber and is positioned on a first side of the graphical representation of the vehicle and a second numerical indication of the at least one numerical indication is associated with the second shock absorber and is positioned on a second side of the graphical representation of the vehicle, the second side of the graphical representation of

the vehicle is opposite the first side of the graphical representation of the vehicle. In another variation thereof, a first graphical indication of the at least one graphical indication is associated with the first shock absorber and is positioned on a first side of the graphical representation of the vehicle and a second graphical indication of the at least one graphical indication is associated with the second shock absorber and is positioned on a second side of the graphical representation of the vehicle, the second side of the graphical representation of the vehicle is opposite the first side of the graphical representation of the vehicle.

[0021] In a further example thereof, the screen layout further a graphical representation of the vehicle, the at least one adjustable shock absorber includes a first shock absorber associated with a first suspension of the at least one suspension and a second shock absorber associated with a second suspension of the at least one suspension, the first suspension being a front suspension of the vehicle and the second suspension being a rear suspension of the vehicle. In a variation thereof, a first numerical indication of the at least one numerical indication is associated with the first shock absorber and is positioned proximate a front portion of the graphical representation of the vehicle and a second numerical indication of the at least one numerical indication is associated with the second shock absorber and is positioned proximate a rear portion of the graphical representation of the vehicle. In another variation thereof, a first graphical indication of the at least one graphical indication is associated with the first shock absorber and is positioned proximate a front portion of the graphical representation of the vehicle and a second graphical indication of the at least one graphical indication is associated with the second shock absorber and is positioned proximate a rear portion of the graphical representation of the vehicle.

[0022] In still a further example thereof, the notification of the active vehicle condition modifier which alters the at least one damping characteristic of the at least one adjustable shock absorber is associated with one of an anti-dive condition, a cornering condition, a ride condition, a squat condition, a brake condition, a roll/cornering condition, and a jump/pitch condition.

[0023] Additional features of the present disclosure will become apparent to those skilled in the art upon consideration of the following detailed description of illustrative embodiments exemplifying the best mode of carrying out the invention as presently perceived.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] The foregoing aspects and many additional features of the present system and method will become more readily appreciated and become better understood by reference to the following detailed description when taken in conjunction with the accompanying drawings.

[0025] FIG. 1 illustrates a representative view of components of a vehicle of the present disclosure having a suspension with a plurality of continuous damping control shock absorbers and a plurality of sensors integrated with a controller of the vehicle;

[0026] FIG. 2 illustrates an adjustable damping shock absorber coupled to a vehicle suspension;

[0027] FIG. 3 illustrates an x-axis, a y-axis, and a z-axis for a vehicle such as an ATV;

[0028] FIG. 4 illustrates a representative view of an exemplary power system for the vehicle of FIG. 1;

[0029] FIG. 5 illustrates a representative view of an exemplary controller of the vehicle of FIG. 1;

[0030] FIG. 6 illustrates a first, perspective view of an exemplary vehicle;

[0031] FIG. 7 illustrates a second, perspective view of the exemplary vehicle of FIG. 6;

[0032] FIG. 8 illustrates a side view of the exemplary vehicle of FIG. 6;

[0033] FIG. 9 illustrates a bottom view of the exemplary vehicle of FIG. 6;

[0034] FIG. 10 illustrates an exemplary representation of an operation of the suspension controller of FIG. 5 and interaction with other vehicle systems;

[0035] FIG. **11** illustrates an exemplary screen layout for a graphical user interface of an operator interface of the vehicle of FIG. **1**;

[0036] FIG. **12** illustrates another exemplary screen layout for the graphical user interface of the operator interface of the vehicle of FIG. **1**;

[0037] FIG. **12A** illustrates a further exemplary screen layout for the graphical user interface of the operator interface of the vehicle of FIG. **1**;

[0038] FIG. **12B** illustrates a further exemplary screen layout for the graphical user interface of the operator interface of the vehicle of FIG. **1**;

[0039] FIG. **12C** illustrates a further exemplary screen layout for the graphical user interface of the operator interface of the vehicle of FIG. **1**;

[0040] FIG. **13** illustrates a representative view of an operation of a suspension controller of FIG. **5** wherein a requested ride mode change is permitted or denied based on a plurality of inputs based on vehicle condition sensors and/or vehicle condition modifier states;

[0041] FIG. **14** illustrates a representative view of an operation of a suspension controller of FIG. **5** wherein a requested ride mode change is permitted or denied based on a plurality of inputs based on vehicle condition sensors and/or vehicle condition modifier states including a current ride mode, a vehicle speed, vehicle acceleration values, and vehicle inertial values;

[0042] FIG. **15** illustrates an exemplary processing sequence of a ride mode change logic of the suspension controller;

[0043] FIG. **16** illustrates an exemplary processing of the controller of FIG. **1** for a power save condition;

[0044] FIG. **17** illustrates an exemplary processing of the controller of FIG. **1** for handling multiple ride mode change requests;

[0045] FIG. **18** illustrates an exemplary aerial view of a terrain, multiple vehicles on a trail of the terrain, and potential ride mode changes based on the terrain or vehicles;

[0046] FIG. **19A** illustrates an exemplary processing sequence of the controller of FIG. **1** related to an airborne detection of the vehicle;

[0047] FIG. **19B** illustrates an exemplary processing sequence of the controller of FIG. **1** related to an airborne detection of the vehicle;

[0048] FIG. **20** illustrates exemplary damping control curves for the adjustable shocks of FIG. **1** in response to an airborne detection of the vehicle;

[0049] FIG. **21** illustrates an exemplary processing sequence of the controller of FIG. **1** regarding airborne detection of the vehicle;

[0050] FIG. **22** illustrates an exemplary processing sequence of the controller of FIG. **1** regarding a steering angle of the vehicle;

[0051] FIG. **23A** illustrates an exemplary processing sequence of the controller of FIG. **1** regarding airborne detection of the vehicle;

[0052] FIG. **23B** illustrates an exemplary processing sequence of the controller of FIG. **1** regarding airborne detection of the vehicle;

[0053] FIG. **24A** illustrates an exemplary processing sequence of the controller of FIG. **1** regarding cornering detection of the vehicle;

[0054] FIG. **24B** illustrates an exemplary processing sequence of the controller of FIG. **1** regarding cornering detection of the vehicle;

[0055] FIG. **25** illustrates exemplary operator inputs of the operator interface of FIG. **1**;

[0056] FIG. **26** illustrates an exemplary terrain sensing embodiment of the controller of FIG. **1**;

[0057] FIG. **27A** illustrates an exemplary terrain progression for the vehicle of FIG. **6**; and

[0058] FIG. **27B** illustrates an exemplary suspension damping level for the terrain progression of FIG. **27A**.

[0059] Corresponding reference characters indicate corresponding parts throughout the several views. Although the drawings represent embodiments of various features and components

according to the present disclosure, the drawings are not necessarily to scale and certain features may be exaggerated in order to better illustrate and explain the present disclosure.

DETAILED DESCRIPTION OF THE DRAWINGS

[0060] For the purposes of promoting an understanding of the principles of the present disclosure, reference will now be made to the embodiments illustrated in the drawings, which are described below. The embodiments disclosed below are not intended to be exhaustive or limited to the precise form disclosed in the following detailed description. Rather, the embodiments are chosen and described so that others skilled in the art may utilize their teachings.

[0061] Referring now to FIG. 1, the present disclosure relates to a vehicle **10** having a suspension system **11** located between a plurality of ground engaging members **12** and a vehicle frame **14**. Exemplary ground engaging members **12** include wheels, skis, guide tracks, treads or other suitable devices for supporting the vehicle relative to the ground. The suspension typically includes springs **16** and shock absorbers **18** coupled between the ground engaging members **12** and the frame **14**. The springs **16** may include, for example, coil springs, leaf springs, air springs or other gas springs. The air or gas springs **16** may be adjustable. See, for example, U.S. Pat. No. 7,950,486, assigned to the current assignee, the entire disclosure of which is incorporated herein by reference.

[0062] The adjustable shock absorbers **18** are often coupled between the vehicle frame **14** and the ground engaging members **12** through an A-arm linkage **70** (See FIG. 2) or other type linkage. Springs **16** are also coupled between the ground engaging members **12** and the vehicle frame **14**. FIG. 2 illustrates an adjustable shock absorber **18** mounted on an A-arm linkage **70** having a first end pivotably coupled to the vehicle frame **14** and a second end pivotably coupled to A-arm linkage **70** which moves with wheel **12**. A damping control activator **74** is coupled to controller **20** by wires **71**. An exemplary damping control activator is an electronically controlled valve which is activated to increase or decrease the damping characteristics of adjustable shock absorber **18**.

[0063] In one embodiment, the adjustable shock absorbers **18** include solenoid valves mounted at the base of the shock body or internal to a damper piston of the shock absorber **18**. The stiffness of the shock is increased or decreased by introducing additional fluid to the interior of the shock absorber, removing fluid from the interior of the shock absorber, and/or increasing or decreasing the case with which fluid can pass from a first side of a damping piston of the shock absorber to a second side of the damping piston of the shock absorber. In another embodiment, the adjustable shock absorbers **18** include a magnetorheological fluid internal to the shock absorber **18**. The stiffness of the shock is increased or decreased by altering a magnetic field experienced by the magnetorheological fluid. Additional details on exemplary adjustable shocks are provided in US Published Patent Application No. 2016/0059660, filed Nov. 6, 2015, titled VEHICLE HAVING SUSPENSION WITH CONTINUOUS DAMPING CONTROL, assigned to the present assignee, the entire disclosure of which is expressly incorporated by reference herein.

[0064] In one embodiment, a spring **16** and shock **18** are located adjacent each of the ground engaging members **12**. In an ATV, for example, a spring **16** and an adjustable shock **18** are provided adjacent each of the four wheels **12**. Some manufacturers offer adjustable springs **16** in the form of either air springs or hydraulic preload rings. These adjustable springs **16** allow the operator to adjust the ride height on the go. However, a majority of ride comfort comes from the damping provided by shock absorbers **18**.

[0065] In an illustrated embodiment, adjustable shocks **18** are electrically controlled shocks for adjusting damping characteristics of shocks **18**. A controller **20** provides signals to adjust damping of the shocks **18** in a continuous or dynamic manner. Adjustable shocks **18** may be adjusted to provide differing compression damping, rebound damping or both. In one embodiment, adjustable shocks **18** include a first controllable valve to adjust compression damping and a second controllable valve to adjust rebound damping. In another embodiment, adjustable shocks include a combination valve which controls both compression damping and rebound damping.

[0066] In an illustrated embodiment of the present disclosure, a user interface **22** is provided in a

location easily accessible to the driver operating the vehicle. Preferably, the user interface **22** is either a separate user interface mounted adjacent the driver's seat on the dashboard or integrated onto a display within the vehicle. User interface **22** includes user input devices to allow the driver or a passenger to manually adjust shock absorber **18** damping during operation of the vehicle based on road conditions that are encountered or to select a preprogrammed active damping profile for shock absorbers **18** by selecting a ride mode. In one embodiment, a selected ride mode alters characteristics of suspension system **11** alone, such as the damping profile for shock absorbers **18**. In one embodiment, a selected ride mode alters characteristics of suspension system **11** and other vehicle systems, such as a driveline torque management system **50**.

[0067] Exemplary input devices for operator interface **22** include levers, buttons, switches, soft keys, and other suitable input devices. Operator interface **22** may also include output devices to communicate information to the operator. Exemplary output devices include lights, displays, audio devices, tactile devices, and other suitable output devices. In another illustrated embodiment, the user input devices are on a steering wheel, handle bar, or other steering control of the vehicle **10** to facilitate actuation of the damping adjustment. A display **24** is also provided on or next to the user interface **22** or integrated into a dashboard display of vehicle **10** to display information related to the shock absorber damping settings.

[0068] Controller **20** receives user inputs from operator interface **22** and adjusts the damping characteristics of the adjustable shocks **18** accordingly. The operator may independently adjust front and rear shock absorbers **18** to adjust the ride characteristics of the vehicle. In certain embodiments, each of the shocks **18** is independently adjustable so that the damping characteristics of the shocks **18** are changed from one side of the vehicle to another. Side-to-side adjustment is desirable during sharp turns or other maneuvers in which different damping profiles for shock absorbers **18** on opposite sides of the vehicle improves the handling characteristics of the vehicle. The damping response of the shock absorbers **18** can be changed in a matter of milliseconds to provide nearly instantaneous changes in damping for potholes, dips in the road, or other driving conditions.

[0069] A plurality of vehicle condition sensors **40** are also coupled to the controller **20**. For example, a global change accelerometer **25** is coupled adjacent each ground engaging member **12**. Each of the accelerometers **25** provides an output signal coupled to controller **20**. Accelerometers **25** provide an output signal indicating movement of the ground engaging members and the suspension components **16** and **18** as the vehicle traverses different terrain.

[0070] Additional vehicle condition sensors **40** may include a vehicle speed sensor **26**, a steering sensor **28**, a chassis supported accelerometer **30**, a chassis supported gyroscope **31**, and other sensors which monitor one or more characteristics of vehicle **10**. Each of vehicle speed sensor **26**, steering sensor **28**, chassis supported accelerometer **30**, and chassis supported gyroscope **31** are operatively coupled to controller **20** and controller **20** receives input from each of vehicle speed sensor **26**, steering sensor **28**, chassis supported accelerometer **30**, and chassis supported gyroscope **31**.

[0071] Accelerometer **30** is illustratively a three-axis accelerometer supported on the chassis to provide an indication of acceleration forces of the vehicle during operation. In one embodiment, accelerometer **30** is located at or close to a center position of vehicle **10**. In the exemplary vehicle **200** illustrated in FIGS. 6-9, accelerometer **30** is located along a longitudinal centerline plane **122** of vehicle **100**. The x-axis, y-axis, and z-axis for a vehicle **10**, illustratively an ATV, are shown in FIG. 3.

[0072] Gyroscope **31** is illustratively a three-axis gyroscope supported on the chassis to provide indications of inertial measurements of the vehicle during operation. In one embodiment, accelerometer **30** is not located at a center of gravity of vehicle **10** and the readings of gyroscope **31** are used by controller **20** to determine the acceleration values of vehicle **10** at the center of gravity of vehicle **10**. In one embodiment, accelerometer **30** and gyroscope **31** are integrated into

suspension controller **86**.

[0073] Additional vehicle condition sensors **40** are disclosed herein and illustratively include in FIG. **1** a brake sensor **32**, a throttle position sensor **34**, a wheel speed sensor **36**, and a gear selection sensor **38**. Each of these vehicle condition sensors **40** are operatively coupled to controller **20** to provide an output signal coupled to controller **20**.

[0074] Referring to FIG. **4**, one embodiment of a driveline torque management system **50** of vehicle **10** is illustrated. Driveline torque management system **50** controls the amount of torque exerted by each of ground engaging members **12**. Driveline torque management system **50** provides a positive torque to one or more of ground engaging members **12** to power the movement of vehicle **10** through a power system **60**. Driveline torque management system **50** further provides a negative torque to one or more of ground engaging members **12** to slow or stop a movement of vehicle **10** through a braking system **75**. In one example, each of ground engaging members **12** has an associated brake of braking system **75**.

[0075] In the illustrated embodiment of FIG. **4**, power system **60** includes a prime mover **62**. Exemplary prime movers **62** include internal combustion engines, two stroke internal combustion engines, four stroke internal combustion engines, diesel engines, electric motors, hybrid engines, and other suitable sources of motive force. To start the prime mover **62**, a power supply system **64** is provided. The type of power supply system **64** depends on the type of prime mover **62** used. In one embodiment, prime mover **62** is an internal combustion engine and power supply system **64** is one of a pull start system and an electric start system. In one embodiment, prime mover **62** is an electric motor and power supply system **64** is a switch system which electrically couples one or more batteries to the electric motor.

[0076] A transmission **66** is coupled to prime mover **62**. Transmission **66** converts a rotational speed of an output shaft **61** of prime mover **62** to one of a faster rotational speed or a slower rotational speed of an output shaft **63** of transmission **66**. It is contemplated that transmission **66** may additionally rotate output shaft **63** at the same speed as output shaft **61**.

[0077] In the illustrated embodiment, transmission **66** includes a shiftable transmission **68** and a continuously variable transmission (“CVT”) **70**. In one example, an input member of CVT **70** is coupled to prime mover **62**. An input member of shiftable transmission **68** is in turn coupled to an output member of CVT **70**. In one embodiment, shiftable transmission **68** includes a forward high setting, a forward low setting, a neutral setting, a park setting, and a reverse setting. The power communicated from prime mover **62** to CVT **70** is provided to a drive member of CVT **70**. The drive member in turn provides power to a driven member through a belt. Exemplary CVTs are disclosed in U.S. Pat. Nos. 3,861,229; 6,176,796; 6,120,399; 6,860,826; and 6,938,508, the disclosures of which are expressly incorporated by reference herein. The driven member provides power to an input shaft of shiftable transmission **68**. Although transmission **66** is illustrated as including both shiftable transmission **68** and CVT **70**, transmission **66** may include only one of shiftable transmission **68** and CVT **70**. Further, transmission **66** may include one or more additional components.

[0078] Transmission **66** is further coupled to at least one differential **72** which is in turn coupled to at least one ground engaging members **12**. Differential **72** may communicate the power from transmission **66** to one of ground engaging members **12** or multiple ground engaging members **12**. In an ATV embodiment, one or both of a front differential and a rear differential are provided. The front differential powering at least one of two front wheels of the ATV and the rear differential powering at least one of two rear wheels of the ATV. In a side-by-side vehicle embodiment having seating for at least an operator and a passenger in a side-by-side configuration, one or both of a front differential and a rear differential are provided. The front differential powering at least one of two front wheels of the side-by-side vehicle and the rear differential powering at least one of multiple rear wheels of the side-by-side vehicle. In one example, the side-by-side vehicle has three axles and a differential is provided for each axle. An exemplary side-by-side vehicle **200** is

illustrated in FIGS. 6-9.

[0079] In one embodiment, braking system **75** includes anti-lock brakes. In one embodiment, braking system **75** includes active descent control and/or engine braking. In one embodiment, braking system **75** includes a brake and in some embodiments a separate parking brake. Braking system **75** may be coupled to any of prime mover **62**, transmission **66**, differential **72**, and ground engaging members **12** or the connecting drive members therebetween. Brake sensor **32**, in one example, monitors when braking system **75** is applied. In one example, brake sensor **32** monitors when a user actuatable brake input, such as brake pedal **232** (see FIG. 7) in vehicle **100**, is applied.

[0080] Referring to FIG. 5, controller **20** has at least one associated memory **76**. Controller **20** provides the electronic control of the various components of vehicle **10**. Further, controller **20** is operatively coupled to a plurality of vehicle condition sensors **40** which monitor various parameters of vehicle **10** or the environment surrounding vehicle **10**. Controller **20** performs certain operations to control one or more subsystems of other vehicle components. In certain embodiments, the controller **20** forms a portion of a processing subsystem including one or more computing devices having memory, processing, and communication hardware. Controller **20** may be a single device or a distributed device, and the functions of the controller **20** may be performed by hardware and/or as computer instructions on a non-transitory computer readable storage medium, such as memory **76**.

[0081] As illustrated in the embodiment of FIG. 5, controller **20** is represented as including several controllers. These controllers may each be single devices or distributed devices or one or more of these controllers may together be part of a single device or distributed device. The functions of these controllers may be performed by hardware and/or as computer instructions on a non-transitory computer readable storage medium, such as memory **76**.

[0082] In one embodiment, controller **20** includes at least two separate controllers which communicate over a network **78**. In one embodiment, network **78** is a CAN network. Details regarding an exemplary CAN network are disclosed in U.S. patent application Ser. No. 11/218,163, filed Sep. 1, 2005, the disclosure of which is expressly incorporated by reference herein. Of course any suitable type of network or data bus may be used in place of the CAN network. In one embodiment, two wire serial communication is used for some connections.

[0083] Referring to FIG. 5, controller **20** includes an operator interface controller **80** which controls communication with an operator through operator interface **22**. A prime mover controller **82** controls the operation of prime mover **62**. A transmission controller **84** controls the operation of transmission system **66**.

[0084] A suspension controller **86** controls adjustable portions of suspension system **11**. Exemplary adjustable components include adjustable shocks **18**, adjustable springs **16**, and/or configurable stabilizer bars. Additional details regarding adjustable shocks, adjustable springs, and configurable stabilizer bars is provided in US Published Patent Application No. 2016/0059660, filed Nov. 6, 2015, titled VEHICLE HAVING SUSPENSION WITH CONTINUOUS DAMPING CONTROL, assigned to the present assignee, the entire disclosure of which is expressly incorporated by reference herein.

[0085] A communication controller **88** controls communications between a communication system **90** of vehicle **10** and remote devices **500** (see FIG. 17). Exemplary remote devices include other vehicles **10'**, personal computing devices **502**, such as cellphones or tablets, a centralized computer system **504** maintaining one or more databases **506**, and other types of devices remote from vehicle **10** or carried by riders of vehicle **10**. In one embodiment, communication controller **88** of vehicle **10** communicates with paired devices over a wireless network. An exemplary wireless network is a radio frequency network utilizing a BLUETOOTH protocol. In this example, communication system **90** includes a radio frequency antenna. Communication controller **88** controls the pairing of devices to vehicle **10** and the communications between vehicle **10** and the remote device. In one embodiment, communication controller **88** of vehicle **10** communicates with remote devices over a cellular network. In this example, communication system **90** includes a cellular antenna and

communication controller **88** receives and sends cellular messages from and to the cellular network. In one embodiment, communication controller **88** of vehicle **10** communicates with remote devices over a satellite network. In this example, communication system **90** includes a satellite antenna and communication controller **88** receives and sends messages from and to the satellite network. In one embodiment, vehicle **10** is able to communicate with other vehicles **10** over a Radio Frequency mesh network and communication controller **88** and communication system **90** are configured to enable communication over the mesh network. An exemplary vehicle communication system is disclosed in U.S. patent application Ser. No. 15/262,113, filed Sep. 12, 2016, titled VEHICLE TO VEHICLE COMMUNICATIONS DEVICE AND METHODS FOR RECREATIONAL VEHICLES, the entire disclosure of which is expressly incorporated by reference herein.

[0086] A steering controller **102** controls portions of a steering system **104**. In one embodiment, steering system **104** is a power steering system and includes one or more steering sensors **28**. Exemplary sensors and electronic power steering units are provided in U.S. patent application Ser. No. 12/135,107, assigned to the assignee of the present application, titled VEHICLE, docket PLR-06-22542.02P, the disclosure of which is expressly incorporated by reference herein.

[0087] A vehicle controller **92** controls lights, loads, accessories, chassis level functions, and other vehicle functions.

[0088] A ride height controller **96** controls the preload and operational height of the vehicle. In one embodiment, ride height controller controls springs **16** to adjust a ride height of vehicle **10**, either directly or through suspension controller **86**. In one example, ride height controller **96** provides more ground clearance in the comfort ride mode compared to the sport ride mode.

[0089] An agility controller **100** controls a braking system of vehicle **10** and the stability of vehicle **10**. Control methods of agility controller **100** may include integration into braking circuits (ABS) such that a stability control system can improve dynamic response (vehicle handling and stability) by modifying the shock damping in conjunction with electronic braking control.

[0090] In one embodiment, controller **20** either includes or is operatively coupled over network **78** to a location determiner **110** which determines a current location of vehicle **10**. An exemplary location determiner **110** is a GPS unit which determines the position of vehicle **10** based on interaction with a global satellite system.

[0091] Referring to FIGS. **6-9**, an exemplary side-by-side vehicle **200** is illustrated. Vehicle **200**, as illustrated, includes a plurality of ground engaging members **12**. Illustratively, ground engaging members **12** are wheels **204** and associated tires **206**. As mentioned herein one or more of ground engaging members **12** are operatively coupled to power system **60** (see FIG. **4**) to power the movement of vehicle **200** and braking system **75** to slow movement of vehicle **200**.

[0092] Referring to the illustrated embodiment in FIG. **6**, a first set of wheels, one on each side of vehicle **200**, generally correspond to a front axle **208**. A second set of wheels, one on each side of vehicle **200**, generally correspond to a rear axle **210**. Although each of front axle **208** and rear axle **210** are shown having a single ground engaging member **12** on each side, multiple ground engaging members **12** may be included on each side of the respective front axle **208** and rear axle **210**. As configured in FIG. **6**, vehicle **200** is a four wheel, two axle vehicle.

[0093] Referring to FIG. **9**, wheels **204** of front axle **208** are coupled to a frame **212** of vehicle **200** through front independent suspensions **214**. Front independent suspensions **214** in the illustrated embodiment are double A-arm suspensions. Other types of suspensions systems may be used for front independent suspensions **214**. The wheels **204** of rear axle **210** are coupled to frame **212** of vehicle **200** through rear independent suspensions **216**. Other types of suspensions systems may be used for rear independent suspensions **216**.

[0094] Returning to FIG. **6**, vehicle **200** includes a cargo carrying portion **250**. Cargo carrying portion **250** is positioned rearward of an operator area **222**. Operator area **222** includes seating **224** and a plurality of operator controls. In the illustrated embodiment, seating **224** includes a pair of

bucket seats. In one embodiment, seating **224** is a bench seat. In one embodiment, seating **224** includes multiple rows of seats, either bucket seats or bench seats or a combination thereof. Exemplary operator controls include a steering wheel **226**, a gear selector **228**, an accelerator pedal **230** (see FIG. 7), and a brake pedal **232** (see FIG. 7). Steering wheel **226** is operatively coupled to the wheels of front axle **208** to control the orientation of the wheels relative to frame **212**. Gear selector **228** is operatively coupled to the shiftable transmission **68** to select a gear of the shiftable transmission **68**. Exemplary gears include one or more forward gears, one or more reverse gears, and a park setting. Accelerator pedal **230** is operatively coupled to prime mover **62** to control the speed of vehicle **200**. Brake pedal **232** is operatively coupled to brake units associated with one or more of wheels **204** to slow the speed of vehicle **200**.

[0095] Operator area **222** is protected with a roll cage **240**. Referring to FIG. 6, side protection members **242** are provided on both the operator side of vehicle **200** and the passenger side of vehicle **200**. In the illustrated embodiment, side protection members **262** are each a unitary tubular member.

[0096] In the illustrated embodiment, cargo carrying portion **250** includes a cargo bed **234** having a floor **256** and a plurality of upstanding walls. Floor **256** may be flat, contoured, and/or comprised of several sections. Portions of cargo carrying portion **250** also include mounts **258** which receive an expansion retainer (not shown). The expansion retainers which may couple various accessories to cargo carrying portion **250**. Additional details of such mounts and expansion retainers are provided in U.S. Pat. No. 7,055,454, to Whiting et al., filed Jul. 13, 2004, titled "Vehicle Expansion Retainers," the entire disclosure of which is expressly incorporated by reference herein.

[0097] Front suspensions **214A** and **214B** each include a shock absorber **260**, respectively. Similarly, rear suspensions **216A** and **216B** each include a shock absorber **262**. In one embodiment each of shock absorbers **260** and shock absorbers **262** are electronically adjustable shocks **18** which are controlled by a controller **20** of vehicle **200**.

[0098] Additional details regarding vehicle **200** are provided in U.S. Pat. Nos. 8,827,019 and 9,211,924, assigned to the present assignee, the entire disclosures of which are expressly incorporated by reference herein. Other exemplary recreational vehicles include ATVs, utility vehicles, snowmobiles, other recreational vehicles designed for off-road use, on-road motorcycles, and other suitable vehicles.

[0099] Referring to FIG. 10, an exemplary control system **300** for controlling the damping of shock absorbers **18** is provided. Suspension controller **86** is operatively coupled to shock absorbers **18** and controls the damping of shock absorbers **18** based on a plurality of inputs. Exemplary inputs are provided in FIG. 10 and throughout this disclosure. Further, additional exemplary inputs for suspension controller **86** and control processing sequences for suspension controller **86** are provided in US Published Patent Application No. 2016/0059660, filed Nov. 6, 2015, titled VEHICLE HAVING SUSPENSION WITH CONTINUOUS DAMPING CONTROL, assigned to the present assignee, the entire disclosure of which is expressly incorporated by reference herein.

[0100] With reference to FIG. 10, steering controller **102** is operatively coupled to a power steering system **302**. Steering controller **102** communicates over network **78** values for one or more monitored parameters to suspension controller **86**. Exemplary monitored parameters include a steering position, for example a steering angle of steering wheel **226** for vehicle **200**, a steering torque, and a steering velocity.

[0101] In some instances, the steering angle of vehicle **100** provided by steering controller **102** may drift over time compared to the actual orientation of the steerable wheels **204** of vehicle **200**, illustratively wheels **204** of front axle **208**. For example, an operator of vehicle **200** may have to turn a steering device, such as steering wheel **226** for vehicle **200**, from its straight line position to maintain vehicle **200** traveling straight along the x-axis (see FIG. 3). As such, steering controller **102** may indicate a slight turn when in reality vehicle **200** is traveling straight. Exemplary reasons for the difference include changes in vehicle **200** as vehicle **200** is broken in and hardware issues,

such as a flat or underinflated tire, a bent steering component, or a bent suspension component.

[0102] As mentioned herein, suspension controller **86** may determine vehicle **10** is cornering based on the steering position, such as steering angle, received from steering controller **102**. In one embodiment, suspension controller **86** receives the steering position from steering controller **102** and based on inputs from other sensors of the plurality of sensors **40** determines a revised steering position to use as the input for determining if vehicle **10** is cornering.

[0103] Referring to FIG. **22**, an exemplary processing sequence **850** of the logic of suspension controller **86** for determining a revised steering position is illustrated. A steering angle value is received by suspension controller **86** from steering system **104**, as represented by block **852**. An updated steering angle value is determined by suspension controller **86**, as represented by block **854**, by applying an offset to the steering angle value. The offset may be applied to the steering angle value in one example by subtracting the offset value and in another example by adding the offset value. The value of the offset is calibrated and stored in a memory accessible by suspension controller **86**. In one embodiment, the offset is not actively adapting when a Steering Angle received over the vehicle network **78** is not valid, an Engine Speed received over the vehicle network **78** is not valid, a Vehicle Speed received over the vehicle network **78** is not valid, and/or a y-axis acceleration is not valid, the y-axis acceleration is not valid when it indicates that vehicle **10** is side hilling. The steering angle adaptation processing sequence **850** is also not active when vehicle **10** is determined to be airborne, braking, or accelerating.

[0104] In one example, the offset value is calibrated by the operator of vehicle **10** during operation of vehicle **10**. The operator positions the steering wheel **226** of vehicle **10** such that vehicle **10** is traveling straight. The operator selects a steering angle calibrate input of operator input **22**.

Suspension controller **86** receives the steering angle value from steering system **104** over a period of time and sets the offset equal to the average steering angle value over that period of time. Once set, the operator deselects the calibrate input or suspension controller **86** exits the calibrate routine.

[0105] In another example, the offset value is calibrated by suspension controller **86** based on a plurality of inputs from the plurality of sensors **40**. Suspension controller **86** determines a direction of travel of vehicle **10** based on the x-axis and y-axis of accelerometer **30** and/or the rotation rate about the z-axis of gyroscope **31**. If suspension controller **86** determines vehicle **10** is traveling straight in the x-axis direction, then vehicle **10** adjusts the offset value in block **854** over time.

[0106] In one embodiment, suspension controller **86** may determine vehicle **10** is cornering based on a Yaw characteristic of vehicle **10**. Referring to FIG. **2**, when vehicle **10** is cornering, vehicle **10** experiences a rotation about the z-axis. By using a Yaw characteristic, such as rotation about the z-axis, suspension controller **86** may account for cornering events that are not initiated using steering angle or cornering events that are inconsistent with steering angle. Based on the terrain, cornering events are sometimes initiated through a braking of vehicle **10** or an acceleration of vehicle **10**.

Further, in some scenarios an operator of vehicle **10** will provide a steering angle input of a cornering event to the right to keep vehicle **10** in a cornering event to the left. The Yaw characteristic based cornering processing sequence is able to more accurately detect cornering events in conditions such as over-steer sliding initiated by low friction surfaces, oversteer sliding initiated by braking, oversteer sliding initiated by acceleration, general maneuvers on ice/snow, and any driving situation with counter steer.

[0107] Referring to FIG. **24A**, an exemplary processing sequence **790** of suspension controller **86** is shown. Suspension controller **86** receives a first sensor input related to an angular movement of the vehicle, as represented by block **792**. In one example, the angular movement of the vehicle is an overall angular movement of the vehicle about the z-axis (see FIG. **2**). In one example, the first sensor is a yaw rate sensor which measures a rotation rate about the z-axis of the vehicle (see FIG. **2**).

[0108] Suspension controller **86** compares the first sensor input to a first threshold, as represented by block **794**. In one example suspension controller **86** compares a magnitude of the first sensor

input to the first threshold. In one example, the magnitude of the first sensor value is averaged over a period of time and the averaged value is compared to the first threshold. If the magnitude of the first sensor input exceeds the first threshold, suspension controller **86** determines that a cornering event is occurring, as represented by block **798**. Otherwise, suspension controller **86** determines that a cornering event has not occurred, as represented by block **796**. If a cornering event has been determined to be occurring, the direction of the cornering event is determined, as represented by block **800**. In one example, the sign of the yaw rotation rate indicates the direction of the cornering event.

[0109] In one embodiment, suspension controller **86** further determines if vehicle **10** is spinning by comparing an angular characteristic of the vehicle, such as the yaw rotation rate, to a second threshold. If the magnitude of the yaw rotation rate exceeds the second threshold, then suspension controller **86** determines that vehicle **10** is spinning. In one example, the angular characteristic is the yaw rotation rate and the second threshold is about 100 degrees per second. In another example, the angular characteristic is a change in the direction of travel of the vehicle of at least 90 degrees over a period of time.

[0110] In one embodiment, when suspension controller **86** determines vehicle **10** is cornering to the left the stiffness of the damping characteristic of the adjustable shocks **18** on the right side of vehicle **10** (passenger side) is increased, when suspension controller **86** determines vehicle **10** is cornering to the right the stiffness of the damping characteristic of the adjustable shocks **18** on the left side of vehicle **10** (driver side) is increased, and when suspension controller **86** determines vehicle **10** is spinning the stiffness of all of the adjustable shocks **18** is increased.

[0111] Referring to FIG. **24B**, another exemplary processing sequence **810** of suspension controller **86** is shown. Suspension controller **86** receives a first sensor input related to an angular movement of the vehicle and a second sensor input related to a position of a steering device, such as steering wheel **226**, of vehicle **200**, as represented by block **812**. In one example, the angular movement of the vehicle is an overall angular movement of the vehicle about the z-axis (see FIG. **2**). In one example, the first sensor is a yaw rate sensor which measures a rotation rate about the z-axis of the vehicle (see FIG. **2**) and the second sensor measures an angle of the steering wheel **226**.

[0112] Suspension controller **86** compares the second sensor input to a second threshold, as represented by block **814**. In one example suspension controller **86** compares a magnitude of the second sensor input to the second threshold. If the magnitude of the second sensor input exceeds the second threshold, suspension controller **86** determines that a cornering event is occurring, as represented by block **816**. Otherwise, suspension controller **86** compares the first sensor input to a first threshold, as represented by block **820**. In one example suspension controller **86** compares a magnitude of the first sensor input to the first threshold. If the magnitude of the first sensor input exceeds the first threshold, suspension controller **86** determines that a cornering event is occurring, as represented by block **816**. Otherwise, suspension controller **86** determines that a cornering event has not occurred, as represented by block **822**. If a cornering event has been determined to be occurring, the direction of the cornering event is determined, as represented by block **818**. In one example, the sign of the yaw rotation rate indicates the direction of the cornering event. In this manner, the angular characteristic of the overall vehicle, for example the yaw rotation rate, takes priority over the angular characteristic of a steering device of the vehicle, for example the angle of the steering wheel **226**. If the steering system indicates the vehicle is turning right and the yaw rate indicates the vehicle is turning left, suspension controller **86** will use the determination of yaw rate sensor to establish the direction of cornering.

[0113] In one embodiment, the yaw angular rate threshold changes as a function of vehicle speed.

[0114] Suspension controller **86** is further operatively coupled to prime mover controller **82**, transmission controller **84**, communication controller **88**, vehicle controller **92**, ride height controller **96**, and agility controller **100**. One or more of prime mover controller **82**, transmission controller **84**, communication controller **88**, vehicle controller **92**, ride height controller **96**, and

agility controller **100** communicates over network **78** values for one or more monitored parameters to suspension controller **86**. Exemplary monitored parameters include a vehicle speed, an engine speed, a brake status (brake pedal **232** applied or not applied), a throttle pedal position (accelerator pedal **230** position), a battery voltage of a battery **80** (powers start-up of vehicle **10**), and any fault information.

[0115] Suspension controller **86** is further operatively coupled to operator interface controller **80** of user interface **22**. Operator interface controller **80**, in one embodiment, communicates over network **78** values for parameters selected by an operator of vehicle **10**. Exemplary parameters include a number of operators or riders, a ride mode selection for suspension system **11**, and other suitable tuning selections for shock absorbers **18** of suspension system **11**. In one example, the operator may input a value for the number of riders and a value estimating a cargo weight through display **24**. In one embodiment, vehicle **200** includes sensors associated with seating **224** which provide an indication of whether a rider is positioned in the respective seats or not. Further, vehicle **200** may include additional sensors which monitor an amount of cargo being carried by cargo carrying portion **250**. Suspension controller **86** also communicates information to operator interface controller **80**. Exemplary information includes a current shock absorber state for shock absorbers **18**, a conditional shock absorber state for shock absorbers **18** (for example, "Turning", "Airborne", and "Anti-Dive"), a ride mode selection for suspension system **11**, fault information, and vehicle metrics. Exemplary vehicle metrics include values of parameters received by suspension controller **86**. In the illustrated embodiment, operator interface controller **80** is operatively coupled to display **24**. In one embodiment, display **24** is a touch screen display that presents one or more screens of a graphical user interface **310** to an operator of vehicle **10**.

[0116] As shown in FIG. **10**, in one embodiment, suspension controller **86** is further operatively coupled to a suspension control rocker switch **312**. Rocker switch **312** is connected through a serial connection to suspension controller **86**. Rocker switch **312** has three positions. Suspension controller **86** determines whether rocker switch **312** is in a first position corresponding to the suspension being in a comfort mode (described herein), a second position corresponding to the suspension being in a sport mode (described herein), and a third position corresponding to the suspension being in a firm mode (described herein).

[0117] Rocker switch **312** may be used in place of display **24** or in addition to the inputs provided through display **24**. In one example, rocker switch **312** selects between multiple ride modes and the inputs of graphical user interface **310** provide operator customization of the vehicle characteristics associated with each of the selectable ride modes as indicated by the configuration/tuning label in FIG. **10**. In another example, vehicle **10** does not include rocker switch **312** and the selection of various ride modes and operator customization of the vehicle characteristics associated with each of the selectable ride modes is received by controller **20** through graphical user interface **310** of display **24**.

[0118] As mentioned in connection with FIG. **10**, in one embodiment, vehicle **10** includes a graphical user interface **310** presented on display **24**. Referring to FIG. **11**, an exemplary screen **320** of graphical user interface **310** is illustrated. Screen **320** includes a first portion **322** having operator selectable inputs **324** to select a ride mode for suspension system **11** and operator selectable inputs **326** to select the manner in which suspension system **11** is adjusted.

[0119] Operator selectable inputs **324** include a plurality of inputs which permit the operator to select from a plurality of predefined ride modes. Illustratively, exemplary inputs **328**, **330**, and **332** correspond to a comfort ride mode, a sport ride mode, and a firm ride mode, respectively. Suspension controller **86** has stored damping profiles that correspond to each ride mode. In the following description, inputs **328**, **330**, and **332** are used to select the various ride modes. However, in one embodiment, the portions of display **24** corresponding to inputs **328**, **330**, and **332** only provide visual feedback to the operator of the current ride mode and are not selectable inputs. Rather, the ride modes are selected through suspension control rocker switch **312**.

[0120] In one embodiment, a comfort ride mode is selected by touching the portion of display **24** at which the comfort button **328** is displayed. A comfort ride mode is generally optimized for comfort and performance. The suspension remains normally soft unless dynamic vehicle conditions sensed by one or more of vehicle condition sensors **40** demand a more firm setting. In one embodiment, a comfort ride mode is selected by actuation of suspension control rocker switch **312** and the portion of display **24** at which the comfort button **328** is displayed is highlighted or otherwise changed in appearance to provide visual feedback to the operator of the selection of the comfort ride mode. In one embodiment, in the comfort mode, suspension controller **86** controls both a compression damping of the adjustable shocks **18** and a rebound damping of the adjustable shocks. Adjustable shocks **18** are set to have a lower compression damping and a lower rebound damping compared to a sport mode. By having a lower rebound damping the suspension will allow the ground engaging members **12** to drop out faster to the terrain. This will create a more comfortable ride for vehicle **10**. In one example, the amount of compression damping, rebound damping, or both is dependent on a ground speed of vehicle **10**.

[0121] A sport ride mode is selected by touching the portion of display **24** at which the sport button **330** is displayed. The sport ride mode, like other ride modes, may be selected through other inputs of operator interface **22**, such as buttons, toggle switches, and other suitable input devices. A sport ride mode increases the baseline damping of shock absorbers **18** compared to the comfort ride mode, more aggressively controls body roll for vehicle conditions such as turning or airborne, and has different speed sensitivity characteristics for increasing the damping of shock absorbers **18**. In one embodiment, a sport ride mode is selected by actuation of suspension control rocker switch **312** and the portion of display **24** at which the sport button **330** is displayed is highlighted or otherwise changed in appearance to provide visual feedback to the operator of the selection of the sport ride mode. In one embodiment, in the sport mode, suspension controller **86** controls both a compression damping of the adjustable shocks **18** and a rebound damping of the adjustable shocks. Adjustable shocks **18** are set to have a higher compression damping and a higher rebound damping compared to a comfort mode. In one example, the amount of compression damping, rebound damping, or both is dependent on a ground speed of vehicle **10**.

[0122] A firm ride mode is selected by touching the portion of display **24** at which the firm button **332** is displayed. A firm ride mode increases the baseline damping of shock absorbers **18** compared to sports mode. In one example, the firm ride mode provides a maximum damping characteristic of shock absorbers **18**. In one embodiment, a firm ride mode is selected by actuation of suspension control rocker switch **312** and the portion of display **24** at which the comfort button **332** is displayed is highlighted or otherwise changed in appearance to provide visual feedback to the operator of the selection of the firm ride mode.

[0123] Below operator selectable inputs **324** is an information panel that provides a textual description of the ride mode selected. Additional ride modes are disclosed in US Published Patent Application No. 2016/0059660 as drive modes, the entire disclosure of which is expressly incorporated by reference herein.

[0124] Operator selectable inputs **326** include a plurality of inputs which permit the operator to select an active suspension management and a passive suspension management. A passive suspension management is selected by touching the portion of display **24** at which a passive button **334** is displayed. In the passive suspension management, suspension controller **86** adjusts the damping profiles of shock absorbers **18** based on the position of suspension control rocker switch **312**. An active suspension management is selected by touching the portion of display **24** at which an active button **336** is displayed. In the active suspension management, suspension controller **86** adjusts the damping profiles of shock absorbers **18** based on logic contained in controller **20** and based on inputs provided through graphical user interface **310** and suspension control rocker switch **312**.

[0125] Operator selectable inputs **326** further includes a second portion **350** which provides a

graphical feedback to the operator of a steering angle of steering wheel **226** through a rotation orientation of icon **352** and of the firmness of each of shock absorbers **18** through a numerical indicator **354A-D** and a graphical indicator **356A-D**. A firmness scale is provided below first portion **322** and second portion **350** which provides a graphical representation of a shock firmness from soft on the left to hard on the right. In one embodiment, the graphical representation changes color from green to yellow to red as the firmness increases. In one example, numerical indicators **354A-D** and graphical indicators **356A-D** provide actual damping values. In another example, numerical indicators **354A-D** and graphical indicators **356A-D** provide smoothed damping values, such as an average damping value over a time window. In one example, an operator input permits the operator to select whether to display actual damping values or smoothed damping values.

[0126] In one embodiment, each of shock absorbers **18** may have individually set damping profiles which may be set by an operator through graphical user interface **310** of **24** or other suitable operator inputs of user interface **22**. In one embodiment, operator inputs of operator interface **22** include at least one input which provides the operator the ability to add a preset damping amount to each of the adjustable shocks **18** (compression, rebound, or both) or to subtract a preset damping amount to each of the adjustable shocks **18** (compression, rebound, or both). In one example, the added or subtracted offset is applied to vehicle speed based damping profiles only. In another example, the added or subtracted offset is applied to one or more selected semi-active modes, such as Cornering, Anti-dive, Anti-squat, Airborne, and other suitable modes.

[0127] In one embodiment, controller **20** may limit power system **60** to maintain vehicle **10** below a set speed, based on the damping parameters of adjustable shocks **10**, to ensure suitable vehicle handling and stability. In one embodiment, controller **20** may store damping profiles for each of a plurality of operators which are selectable through graphical user interface **310** of display **24**.

[0128] Referring to FIG. **12**, an exemplary screen **370** of graphical user interface **310** is illustrated. Screen **370** includes a first portion **372** having operator selectable inputs **374** to select a ride mode for suspension system **11** and operator selectable inputs **376** to select a setting for transmission **66**.

[0129] Operator selectable inputs **374** includes a plurality of inputs which permit the operator to select from a plurality of predefined ride modes. Illustratively, exemplary inputs **378**, **380**, and **382** correspond to a trail ride mode, a sport ride mode, and a race ride mode, respectively. Below operator selectable inputs **324** is an information panel that provides a textual description of the ride mode selected. Suspension controller **86** has stored damping profiles that correspond to each ride mode. In the following description, inputs **378**, **380**, and **382** are used to select the various ride modes. However, in one embodiment, the portions of display **24** corresponding to inputs **378**, **380**, and **382** only provide visual feedback to the operator of the current ride mode and are not selectable inputs. Rather, the ride modes are selected through suspension control rocker switch **312**.

[0130] A trail ride mode is selected by touching the portion of display **24** at which the trail button **378** is displayed. A trail ride mode is generally an aggressive setup of the suspension, transmission, and throttle responses tuned for high performance. In one embodiment, a trail ride mode is selected by actuation of suspension control rocker switch **312** and the portion of display **24** at which the comfort button **378** is displayed is highlighted or otherwise changed in appearance to provide visual feedback to the operator of the selection of the trail ride mode.

[0131] A sport ride mode is selected by touching the portion of display **24** at which the sport button **380** is displayed. A sport ride mode increases the baseline damping of shock absorbers **18** compared to the trail ride mode, more aggressively controls body roll for vehicle conditions such as turning or airborne, and has different speed sensitivity characteristics for increasing the damping of shock absorbers **18**. In one embodiment, a sport ride mode is selected by actuation of suspension control rocker switch **312** and the portion of display **24** at which the sport button **380** is displayed is highlighted or otherwise changed in appearance to provide visual feedback to the operator of the selection of the sport ride mode.

[0132] A race ride mode is selected by touching the portion of display **24** at which the race button

382 is displayed. A race ride mode increases the baseline damping of shock absorbers **18** compared to sports mode. In one example, the race ride mode provides a maximum damping characteristic of shock absorbers **18**. In one embodiment, a race ride mode is selected by actuation of suspension control rocker switch **312** and the portion of display **24** at which the race button **328** is displayed is highlighted or otherwise changed in appearance to provide visual feedback to the operator of the selection of the race ride mode.

[0133] Operator selectable inputs **376** include a plurality of inputs which permit the operator to select a setting for transmission **66**. A manual setting is selected by touching the portion of display **24** at which a manual button **384** is displayed. An automatic setting is selected by touching the portion of display **24** at which an auto button **386** is displayed. In one embodiment, the manual setting for transmission **66** means the operator has control over the operation of CVT **70** and the auto setting for transmission **66** means CVT **70** functions without operator intervention. Exemplary CVTs **70** which are configurable in both a manual setting and an auto setting are provided in U.S. Pat. No. 9,429,235, the entire disclosure of which is expressly incorporated by reference herein.

[0134] Screen **370** further includes a second portion **388** which provides a graphical feedback to the operator of the firmness of each of shock absorbers **18** through a numerical indicator **390A-D** and a graphical indicator **392A-D**. In one example, a height of the solid portion of the graphical indicator represents a level of a damping characteristic, either a compression damping or a rebound damping. In one example, the height is not linearly related to the damping characteristic to provide a visual emphasis. Additionally color variation may be used for visual emphasis (green, yellow, orange, red). Further, second portion **388** provides a graphical feedback of any vehicle condition modifiers which are currently active, as represented by icon **394**. In one example, an operator input permits the operator to select whether to display a active vehicle condition modifier icon **394** or not. In the illustrated example, an anti-dive icon is displayed. In an anti-dive condition, suspension controller **86** in response to an indication of heavy braking from brake sensor **32** adjusts the damping levels of adjustable shocks **18** adjacent the front axle to be firmer to reduce “dive” of the vehicle. In one embodiment, in the anti-dive condition, suspension controller **86** in response to an indication of heavy braking from brake sensor **32** adjusts the compression damping levels of adjustable shocks **18** adjacent the front axle to be firmer to reduce “dive” of the vehicle, increase the rebound damping of the rear adjustable shocks **18** of vehicle **10** to decrease the rebound speed of the adjustable shocks **18** and provide improved pitch control of vehicle **10**, and decrease the rebound damping of the front adjustable shocks **18** of vehicle **10** to allow the ground engaging members **12** to get back to the terrain while braking in chatter (e.g. experiencing a high speed harsh trail). In one example, the amount of compression damping, rebound damping, or both is dependent on a ground speed of vehicle **10**.

[0135] Other exemplary vehicle condition modifiers include the Cornering Condition, the Ride Condition, the Squat Condition, the Brake Condition, the Roll/Cornering Condition, the Jump/Pitch Condition, and other factors or combination of factors that cause changes in the damping level of the adjustable shocks in US Published Patent Application No. 2016/0059660, filed Nov. 6, 2015, titled VEHICLE HAVING SUSPENSION WITH CONTINUOUS DAMPING CONTROL, assigned to the present assignee, the entire disclosure of which is expressly incorporated by reference herein. In one embodiment, operator interface **22** includes operator inputs whereby an operator may select one or more of the vehicle condition modifiers to deactivate or activate. Deactivated vehicle condition modifiers will not be used by suspension controller **86** in adjusting the damping characteristics of adjustable shocks **18**.

[0136] In one embodiment, in the Cornering Condition, suspension controller **86** in addition to changes to the compression damping will increase the rebound damping on adjustable shocks **18** which are inside on the turn, such as the left side adjustable shocks for a left turn of vehicle **10**, thereby decreasing the rebound speed of the adjustable shocks **18** and provide improved roll control. In one example, the amount of compression damping, rebound damping, or both is

dependent on a ground speed of vehicle **10**. In one embodiment, in the Squat Condition, in response to an indication of an acceleration event of vehicle **10**, suspension controller **86** will increase the rebound damping of the front adjustable shocks of vehicle **10** resulting in a decrease of the rebound speed of the adjustable shock **18** and an improved pitch control of vehicle **10**. In one example, the amount of compression damping, rebound damping, or both is dependent on a ground speed of vehicle **10**.

[0137] Referring to FIGS. **12A-C**, additional exemplary screens **340**, **342**, and **344** of graphical user interface **310** are illustrated. Each of screens **340**, **342**, and **344** includes a visual feedback representation **346** of a g-force experienced by vehicle **10**. G-force representation **346** includes an indication **348** of a steering angle of steering wheel **226** and an indication **349** of the magnitude (distance from a center of visual feedback representation **346**) and the direction (angular position of indication **349** relative to the center of visual feedback representation **346**) of the g-force experienced. In FIG. **12A**, the g-force is towards the driver rear portion of the vehicle. In one embodiment, the screen layouts disclosed herein may be used as part of a demonstration mode wherein the screen functionality is shown while the vehicle is stationary.

[0138] In one embodiment, suspension controller **86** is microprocessor-based and includes processing instructions stored on a non-transitory computer readable medium, such as memory **76**, which are executable by the microprocessor of suspension controller **86** to control operation of suspension system **11**. Referring to FIG. **13**, suspension controller **86** may execute a ride mode change logic **400** which provides control signals to suspension system **11**, such as the electronically controlled valves of shock absorbers **18**, to achieve various configurations of suspension system **11**, such as race, trail, sport, and other suitable configurations. In the case of multi-operator selectable ride modes, ride mode change logic **400** may permit or deny a change from a current, first ride mode to a requested, second ride mode. The term “logic” as used herein includes software and/or firmware executing on one or more programmable processors, application-specific integrated circuits, field-programmable gate arrays, digital signal processors, hardwired logic, or combinations thereof. Therefore, in accordance with the embodiments, various logic may be implemented in any appropriate fashion and would remain in accordance with the embodiments herein disclosed. A non-transitory machine-readable medium comprising logic can additionally be considered to be embodied within any tangible form of a computer-readable carrier, such as solid-state memory, magnetic disk, and optical disk containing an appropriate set of computer instructions and data structures that would cause a processor to carry out the techniques described herein. This disclosure contemplates other embodiments in which suspension controller **86** is not microprocessor-based, but rather is configured to control operation of suspension system **11** based on one or more sets of hardwired instructions and/or software instructions stored in memory **76**. Further, suspension controller **86** may be contained within a single device or be a plurality of devices networked together to provide the functionality described herein.

[0139] Suspension controller **86** receives multiple inputs that may be used by ride mode logic **400** to determine whether to permit a change from a current, first ride mode to a requested, second ride mode. Suspension controller **86** receives a request **410** to change to a requested, second ride mode. The request **410** is provided by an operator through user interface **22**. In one embodiment, graphical user interface **310** displayed on display **24** is used to select a requested, second ride mode. For example, any one of comfort button **328**, sport button **330**, or firm button **332** of second portion **350** or any one of trail button **378**, sport button **380**, and race button **382** of second portion **388** may be used to select a requested, second ride mode. In one embodiment, a requested, second ride mode is selected by actuation of rocker switch **312**.

[0140] Additionally, suspension controller **86** receives a plurality of inputs **420** based on the operational characteristics of vehicle **10**. Suspension controller **86** receives inputs from a plurality of vehicle condition sensors **40**. The vehicle condition sensors **40** may either actively provide an indication by sending a sensor signal or passively provide an indication by making available a

monitored characteristic, such as a voltage, a temperature, a pressure or other suitable characteristics. Suspension controller **86** further, either receives or initiates based on received inputs from the plurality of vehicle condition sensors **40**, one or more vehicle condition modifier states. Exemplary vehicle condition modifier states are mentioned herein.

[0141] Ride mode change logic **400** determines whether request **410** passes a ride mode change criteria based on the plurality of operational characteristics **420**. If the ride mode change criteria are passed, suspension controller **86** permits the requested ride mode change to occur, as represented by block **422**. For instance, if the current ride mode was a comfort ride mode (selection input **328** in FIG. **11**) and the operator selects a firm ride mode (selection input **332** in FIG. **11**), suspension controller **86** would alter suspension **11** to have characteristics based on the firm ride mode. For example, suspension controller **86** would alter a damping profile of shock absorbers **18** to the default damping profile stored in memory **76** for the firm ride mode. If the ride mode criteria fails, suspension controller **86** denies the requested ride mode change, as represented by block **424**, and suspension system **11** remains in the current selected ride mode until the operating conditions are resolved and/or there has been another switch state change.

[0142] Referring to FIG. **14**, a first exemplary set of operational characteristics **420** includes a current ride mode characteristic **440**, a vehicle speed **442**, one or more acceleration values **444**, and an indication of any active vehicle condition modifier states **446**. In one embodiment, vehicle speed **442** is based on an input from vehicle speed sensor **26**. In one embodiment, the one or more acceleration values **444** include a z-plane acceleration from chassis supported accelerometer **30**, a y-plane acceleration from chassis supported accelerometer **30**, and an x-plane acceleration from chassis supported accelerometer **30**.

[0143] An exemplary processing sequence **450** of ride mode change logic **400** of suspension controller **86** is illustrated in FIG. **15**. Ride mode change logic **400** determines if the requested ride mode is different than the current ride mode, as represented by block **451**. Ride mode change logic **400** determines if the requested ride mode has a firmer damping profile than the current ride mode, as represented by block **452**. The current ride mode is the active damping profile and may either be a previously selected pre-set ride mode or a custom operator specified damping profile. If the requested ride mode is firmer than the current ride mode, ride mode change logic **400** permits the ride mode change, as represented by block **422**. If the requested ride mode is not firmer than the current ride mode, ride mode change logic **400** checks to see if there are any active vehicle condition modifier states, as represented by block **454**. If there are active vehicle condition modifier states, ride mode change logic **400** denies the requested ride mode change, as represented by block **424**. For example, if vehicle **10** is currently cornering as determined based on a steering angle of the steering wheel, ride mode change logic **400** would not permit a change from a firmer damping profile to a softer damping profile.

[0144] If there are not any active vehicle condition modifier states, ride mode change logic **400** reviews the inputs from vehicle condition sensors **40** to determine whether to permit the ride mode change, as represented by block **456**. For example, ride mode change logic **400** may not allow a change to a softer damping profile when a vehicle speed exceeds a vehicle speed threshold. Further, ride mode change logic **400** may not allow a change to a softer damping profile if a x-plane acceleration is greater than a x-plane acceleration threshold or if an absolute value of a y-plane acceleration is greater than a y-plane acceleration threshold. If the inputs from the vehicle sensors are acceptable, ride mode change logic **400** permits the ride mode change as represented by block **422**. Otherwise, ride mode change logic **400** denies the ride mode change, as represented by block **424**. Referring to Table 1, an exemplary set of conditions based on vehicle sensor values that ride mode change logic **400** uses in deciding whether to permit or deny a ride mode change request. In the examples provided in Table 1,

TABLE-US-00001
TABLE 1 Current Requested Ride Mode Ride Mode Conditions Trail Sport No conditions - Sport has a firmer damping profile that Trail Sport Trail X-plane acceleration value

less than X-plane threshold value Absolute value of Y-plane acceleration less than Y-plane threshold value Sport Race No conditions - Race has a firmer damping profile than Sport Race Sport X-plane acceleration value less than X-plane threshold value Absolute value of Y-plane acceleration less than Y-plane threshold value

the Sport ride mode has a firmer damping profile than the Trail ride mode and the Race ride mode has a firmer damping profile than both of the Sport ride mode and the Trail ride mode.

[0145] A first exemplary damping profile may be a constant value for a ride mode. A second damping profile may be a function or a look-up table that provides a damping value based on one or more static or dynamic inputs. For example, a Sport ride mode may have different damping values for shock absorbers **18** based on a first static input, cargo weight, and/or a second static input, number of riders. In one example, the Sport ride mode has a first damping value for vehicle **10** when vehicle **10** has a single rider and no cargo and a second damping value for vehicle **10** when vehicle **10** has two riders and no cargo, the second damping value being firmer than the first damping value. Further, a Sport ride mode may have multiple different damping values for shock absorbers **18** based on a first dynamic input, a detected vehicle speed. In one example, the Sport ride mode has a first damping value for a first vehicle speed and a second, firmer damping value for a second vehicle speed, the second vehicle speed being greater than the first vehicle speed.

[0146] Ride modes are described herein in connection with the settings of suspension system **11**. In one embodiment, a selected ride mode influences other systems of vehicle **10**, such as prime mover **62**, transmission **66**, and other systems of vehicle **10**.

[0147] Damping profiles may be altered during the operation of vehicle **10** beyond the values associated with the current ride mode. For example, if the operation of vehicle **10** results in vehicle **10** being placed in a vehicle condition modifier state, suspension controller **86** will alter the current damping profile based on the vehicle condition modifier state until the vehicle condition modifier state is concluded. In one example, in response to an Anti-Dive modifier state, suspension controller **86** increases the firmness of shock absorbers **18** associated with front axle **208** of vehicle **100**.

[0148] As another example, if based on one or more of vehicle condition sensors **40**, a power save condition is detected, suspension controller **86** may alter the damping profile of shock absorbers **18**. Referring to FIG. **16**, controller **20** executes a processing sequence **480** to monitor for a power save condition. In one example, processing sequence **480** is executed by vehicle controller **92**, but may be executed by any one or more of the controllers of controller **20** or if controller **20** is a single controller handling the functionality of the controllers listed in FIG. **5** then by controller **20**.

[0149] Prime mover controller **82** receives an indication of the voltage of a battery **80** of vehicle **10** from one or more of vehicle condition sensors **40**, an indication of an engine speed from one or more of vehicle condition sensors **40**, and a current ride mode, as represented by block **482**. The measured voltage of the battery is passed through a low pass filter to provide an average measured voltage, as represented by block **484**. Based on the measured voltage value, engine speed, and ride mode, prime mover controller **82** determines if a power load shed is required, as represented by block **486**. Prime mover controller **82** reduces the power consumption of various vehicle systems to reduce the power drain on battery **80**.

[0150] Prime mover controller **82** determines an amount of load shed to be borne by various vehicle systems, as represented by block **488**. In one embodiment, the amount of load shed required by each vehicle system depends on the current ride mode of vehicle **10**. For example, in a first ride mode suspension controller **86** may have constant damping specified for shock absorbers **18** while in a second ride mode suspension controller **86** may have variable damping for shock absorbers **18**. A determined load shed amount or consumption limit is distributed to affected controllers, as represented by block **490**. Each controller, in response to the determined load shed or consumption limit, alters its operation. For example, suspension controller **86** may adjust the damping of shock absorbers **18** to be firmer and not permit softer settings to reduce the number of

adjustments made to shock absorbers **18**. If the consumption limit is below a threshold, suspension controller **86**, in one embodiment, defaults the damping profile of adjustable shocks **18** to be 100% firm. As another example, transmission controller **84** may reduce the number of permitted shifts of transmission **66** in response to the determined load shed or consumption limit.

[0151] Referring to FIG. **17**, vehicle controller **92** may receive multiple ride mode requests from various sources. Vehicle controller **92** may receive a first request **520** from user interface **22** and a second request **522** from communication controller **88**. Communication controller **88** receives second request **522** from one of remote devices **500**. Exemplary remote devices **500** include personal computing devices **502** which may be carried by a rider of vehicle **10** or be spaced apart from vehicle **10**. Further, a second vehicle **10'** may send second request **522** to vehicle **10**. In one embodiment, vehicle **10** and second vehicle **10'** are part of a mesh network and second vehicle **10'** has altered its ride mode and broadcasts that to vehicle **10** or based on the terrain and/or performance of second vehicle **10'** suggestions for preferred ride modes or damping profile adjustments are broadcast to vehicle **10**. Further, a centralized computer system **504** may send second request **522** to vehicle **10**. In one example, any one of personal computing devices **502**, second vehicle **10'**, or centralized computer system **504** broadcasts second request **522** to any vehicle **10** capable of receiving second request **522**. The second request **522** may include location information, such as GPS coordinates, that vehicle controller **92** may use to determine whether second request **522** is pertinent or not based on the current location of vehicle **10** as determined by location determiner **110**.

[0152] Vehicle controller **92** receives both first request **520** and second request **522**. Vehicle controller **92** determines which one of first request **520** and second request **522** to implement, if either, as represented by block **524**. In making this determination, vehicle controller **92** determines whether an auto/manual input **528** of user interface **22** has been set to auto or manual, as represented by block **526**. In manual mode, the operator is controlling the damping profile of shock absorbers **18**. In the auto mode, the operator is permitting other inputs to control the damping profile of shock absorbers **18**. In one embodiment, vehicle controller **92** provides on display **24** an indication of an incoming request from remote devices **500**. The operator may actuate an override input **530** of user interface **22** to override the incoming request from remote devices **500**, as represented by block **532**. In one example, the override input **530** is displayed as part of graphical user interface **310**. Once vehicle controller **92** has determined which of the multiple requests **520**, **522** to implement, vehicle controller **92** communicates the request to the system controllers to implement, illustratively prime mover controller **82**, transmission controller **84**, and suspension controller **86**.

[0153] Referring to FIG. **18**, an aerial view **600** of a representative trail for vehicles **10** is shown. In one embodiment, aerial view **600** is shown on display **24**. The trail starts at location **602**, follows a trail path **604** (dashed line), to a trail end **606**. A lead vehicle **10A** is illustrated along with a follower vehicle **10B**. Lead vehicle **10A** is traversing the trail first and broadcasts recommendations to vehicle **10B** to alter the ride mode, such as altering the damping profile of shock absorbers **18**. The trail may also include discrete locations, illustratively locations **608A-608D**, at which either lead vehicle **10B** or databases **506** accessible by centralized computer system **504**, indicates a desired ride mode, such as a damping profile of shock absorbers **18**. In one embodiment, controller **20** of vehicle **10A** provides visual feedback to the operator of vehicle **10A** of an upcoming ride mode change.

[0154] Referring to FIG. **19A**, a processing sequence **640** related to vehicle alterations to make in light of vehicle **10** being airborne is provided. Controller **20** determines if vehicle **10** is airborne, as represented by block **652**. In one embodiment, controller **20** monitors the output of accelerometer **30** to determine when vehicle **10** is in a freefall. In this manner, accelerometer **30** is a type of airborne sensor in that it is able to provide an indication of when vehicle **10** is airborne.

[0155] In one embodiment, the output of each of the three axes of accelerometer **30** is monitored by

controller **20**. Controller **20** determines that vehicle **10** is airborne in response to a magnitude of a resultant acceleration vector (based on the x-axis component vector, the y-axis component vector, and the z-axis component vector) being below a first threshold value and, optionally in one example, the magnitude of each of the x-axis acceleration and the magnitude of the y-axis acceleration being below a second threshold.

[0156] Referring to FIG. **23A**, an exemplary processing sequence **750** of the logic of controller **20**, such as suspension controller **86**, is shown. Controller **20** receives the acceleration values for the x-axis, the y-axis, and the z-axis from accelerometers **30**, as represented by block **752**. Controller **20** determines the magnitude of the resultant acceleration vector, as represented by block **754**.

Controller **20** compares the magnitude of the resultant acceleration vector to a first threshold, as represented by block **756**. If the magnitude of the resultant acceleration vector is less than the first threshold, controller **20** determines vehicle **10** is experiencing an airborne event, as represented by block **764**. Otherwise, controller **20** determines that vehicle **10** is not experiencing an airborne event, as represented by block **760**, and the next acceleration values are reviewed, as represented by block **752**. In one example, the first threshold value is about 0.3 times the gravitational acceleration on planet Earth (G). As mentioned herein, if accelerometer **30** is not located at the center of gravity of vehicle **10**, the inertial values of gyroscope **31** are used in combination with the raw measurement of acceleration values to determine the equivalent acceleration values at the center of gravity of vehicle **10**.

[0157] In another embodiment, a magnitude of each of the x-axis acceleration component vector, the y-axis acceleration component vector, and the z-axis acceleration component vector is compared to a respective threshold value. Controller **20** determines that vehicle **10** is airborne in response to each of the magnitude of the x-axis acceleration component vector, the magnitude of the y-axis acceleration component vector, and the magnitude of the z-axis acceleration component vector being less than their respective threshold values. By monitoring each acceleration component magnitude individually instead of a composite acceleration magnitude, controller **20** may provide improved airborne detection in high speed jumping maneuvers and better account for wind resistance.

[0158] Referring to FIG. **22B**, an exemplary processing sequence **770** of the logic of controller **20**, such as suspension controller **86**, is shown. Controller **20** receives the acceleration values for the x-axis, the y-axis, and the z-axis from accelerometers **30**, as represented by block **772**. Controller **20** compares the x-axis acceleration value to a first threshold, as represented by block **774**; the y-axis acceleration value to a second threshold, as represented by block **776**; and the z-axis acceleration value to a third threshold, as represented by block **780**. In FIG. **22B**, blocks **774**, **776**, and **780** are illustrated as being executed sequentially. In one embodiment, controller **20** performs the functions of blocks **774**, **776**, and **780** in parallel. Returning to FIG. **22B**, if the x-axis acceleration value is less than the first threshold, controller **20** moves on to block **776**. Otherwise, controller **20** determines that vehicle **10** is not experiencing an airborne event, as represented by block **778**, and the next acceleration values are reviewed, as represented by block **772**. At block **776**, controller **20** compares the y-axis acceleration value to a second threshold. If the magnitude of the y-axis acceleration value is less than the second threshold, controller **20** moves on to block **780**.

Otherwise, controller **20** determines that vehicle **10** is not experiencing an airborne event, as represented by block **778**, and the next acceleration values are reviewed, as represented by block **772**. At block **780**, controller **20** compares the z-axis acceleration value to a third threshold. If the magnitude of the z-axis acceleration value is less than the third threshold, controller **20** determines that vehicle **10** is experiencing an airborne event, as represented by block **782**. Otherwise, controller **20** determines that vehicle **10** is not experiencing an airborne event, as represented by block **778**, and the next acceleration values are reviewed, as represented by block **772**. In one example, the respective threshold values are about 0.3 times the gravitational acceleration on planet Earth (G). In one example, at least two of the first threshold, the second threshold, and the third

threshold have the same value. In another example, each of the first threshold, the second threshold, and the third threshold have different values. As mentioned herein, if accelerometer **30** is not located at the center of gravity of vehicle **10**, the inertial values of gyroscope **31** are used in combination with the raw measurement of acceleration values to determine the equivalent acceleration values at the center of gravity of vehicle **10**.

[0159] Exemplary detection of an airborne event is described in US Published Patent Application No. 2016/0059660, filed Nov. 6, 2015, titled VEHICLE HAVING SUSPENSION WITH CONTINUOUS DAMPING CONTROL, assigned to the present assignee, the entire disclosure of which is expressly incorporated by reference herein. Exemplary detection of an airborne event is described in U.S. Pat. No. 9,381,810, filed Jun. 3, 2011, titled ELECTRONIC THROTTLE CONTROL, assigned to the present assignee, the entire disclosure of which is expressly incorporated by reference herein.

[0160] In response to determining that vehicle **10** is airborne, controller **20** adjusts the damping profile of shock absorbers **18**, as represented by block **654**. In one embodiment, controller **20** increases a pre-load firmness of shock absorbers **18**. In one example, controller **20** increases the firmness of the damping profile of shock absorbers **18** at different rates depending on vehicle characteristics, such as vehicle speed, number of riders, cargo weight, amount of time in the air, and current ride mode. Referring to FIG. **20**, an illustrative example **670** is provided. Once it has been detected that vehicle **10** is airborne, controller **20** increases the damping of shock absorbers **18** over time from a current damping value **672** to a maximum damping value **674** in accordance with the vehicle characteristics. Curve **676** represents a first set of vehicle characteristics, such as a single rider and no cargo, and curve **678** represents a second set of vehicle characteristics, such as multiple riders and cargo. In one example, the damping value is increased in a linear fashion from current damping value **672** over time. In one embodiment, in response to the detection of an Airborne condition, suspension controller **86** increases the compression damping of all of adjustable shocks **18** and decreases the rebound damping of all of the adjustable shocks **18**. This setup will increase the rebound speed of the adjustable shocks **18** and permit the suspensions of vehicle **10** to drop out faster for the next compression hit to the adjustable shocks **18**.

[0161] Referring to FIG. **19B**, a processing sequence **650** related to vehicle alterations to make in light of vehicle **10** being airborne is provided. Controller **20** determines if vehicle **10** is airborne, as represented by block **652**. In response to determining that vehicle **10** is airborne, controller **20** adjusts the damping profile of shock absorbers **18**, as represented by block **654**. Further, in response to determining that vehicle **10** is airborne, controller **20** adjusts the adjustable driveline torque management characteristic of the driveline torque management system **50**, as represented by block **656**.

[0162] Referring to FIG. **21**, a processing sequence **700** related to vehicle alterations made by vehicle controller **92** in light of vehicle **10** being airborne is provided. Vehicle controller **92** receives output from accelerometer **30**, as represented by block **702**, and monitors the acceleration of vehicle **10**. The output is sent through a low pass filter to average the data, as represented by block **704**. Vehicle controller **92** executes a freefall algorithm to determine if vehicle **10** is falling, as represented by block **704**. In one example, vehicle controller **92** determines if the z-plane acceleration is a negative value indicating a downward movement of vehicle **10**. The received data is de-bounced by a delay amount, as represented by block **708**. After the de-bounce delay the data is checked again to determine if the z-plane acceleration is a negative value. The data is continued to be checked until a timer expires, as represented by block **710**. If the timer expires and the z-plane acceleration continues to be negative, vehicle controller **92** determines that vehicle **10** is airborne, as represented by block **712**. In response thereto, vehicle controller **92** or the individual system controllers make adjustments to the vehicle characteristics. If the timer has not expired but a threshold amount of time has passed, controller **92** in response to a positive z-plane acceleration or the detection of gravity determines the vehicle **10** was airborne. The increase in firmness is

controlled by the appropriate curve in FIG. 20. In one embodiment, airborne detection is determined by monitoring the output of each of the three axes of accelerometer **30** with controller **20**. Controller **20** determines that vehicle **10** is airborne in response to a magnitude of a resultant acceleration vector (based on the x-axis component vector, the y-axis component vector, and the z-axis component vector) being above a first threshold value and the magnitude of each of the x-axis acceleration and the magnitude of the y-axis acceleration being below a second threshold.

[0163] Once the vehicle has been determined to be airborne, suspension controller **86** may increase the suspension damping to provide a firmer suspension for landing, as represented by block **714**.

Prime mover controller **82** and transmission controller **84** may take steps to protect the driveline of vehicle **10**, as represented by block **716**. Prime mover controller **82** may limit the torque output of prime mover **62**. Transmission controller **84** may prevent shifting transmission **66** to a higher gear.

[0164] Further, the output of power system **60** may be increased or decreased to change the rotational speed of one or more of ground engaging members **12** and change the attitude of vehicle **10** while vehicle **10** is in the air, as represented by block **718**. In one example, this is accomplished by adjusting an output of prime mover **62**, altering a speed ratio of transmission **66**, or both.

Further, braking system **75** may be activated to slow the rotational speed of one or more of ground engaging members **12** to change the attitude of vehicle **10**. Controller **20** through changes to driveline torque management system **50** may automatically change the pitch of vehicle **10** while vehicle **10** is airborne.

[0165] Controller **20** may further record information regarding airborne events, as represented by block **720**. Exemplary information includes a time duration of each jump, the peak gravitational value experienced by vehicle **10** upon landing, number of jumps in a given period of time, and other information.

[0166] In one embodiment, a position of shock absorbers **18** is used to detect if vehicle **10** is airborne. In one example, if shocks **18** relaxed beyond a calibrated extension value, the vehicle **10** was determined to be airborne. In one variation, a majority of the shocks had to demonstrate a relaxation beyond a calibrated extension value for the vehicle to be considered airborne. Once the shocks compressed beyond a threshold, the dampening profile was adjusted firmer.

[0167] Referring to FIG. 25, in one embodiment, suspension controller **86** is further operatively coupled to a plurality of operable selectable inputs of user interface **22** through which an operator of vehicle **10** may select a terrain ride mode setting for adjustable shocks **18** and/or adjustable springs **16**. The selectable inputs may be switches, buttons, touch regions on display **24**, dials, and other suitable user inputs. In the illustrated embodiment of FIG. 25, a desert ride mode input **900**, an ice ride mode input **902**, a sand ride mode input **904**, and a trail ride mode input **906** are provided. Other suitable terrain ride mode inputs may be provided.

[0168] In one embodiment, selection of desert ride mode input **900** results in suspension controller **86** adjusting adjustable shocks **18** to a desert setup for desert conditions. In one example, the desert setup may be user defined based on prior vehicle performance in desert conditions.

[0169] In one embodiment, selection of ice ride mode input **902** results in suspension controller **86** adjusting adjustable shocks **18** to an ice setup for ice conditions. In one example, the ice setup may be user defined based on prior vehicle performance in ice conditions. For instance, an ice setup may alter cornering detection algorithms to provide different operational thresholds for low friction surfaces.

[0170] In one embodiment, selection of sand ride mode input **904** results in suspension controller **86** adjusting adjustable shocks **18** to a sand setup for sand conditions. In one example, the sand setup may be user defined based on prior vehicle performance in sand conditions.

[0171] In one embodiment, selection of trail ride mode input **906** results in suspension controller **86** adjusting adjustable shocks **18** to a trail setup for trail conditions. In one example, the trail setup may be user defined based on prior vehicle performance in trail conditions.

[0172] Referring to FIG. 26, in one embodiment, suspension controller **86** includes terrain sensing

logic **912** which based on a plurality of inputs from vehicle condition sensors **40** determines a terrain being experienced by vehicle **10** and adjusts the damping characteristics of one or more of adjustable shocks **18** based thereon. Suspension controller **86** may adjust the compression damping characteristic of one or more of adjustable shocks **18**, the rebound damping characteristics of one or more adjustable shocks **18**, and/or both the compression and rebound damping of one or more adjustable shocks **18**.

[0173] As shown in FIG. **26**, terrain sensing logic **912** receives a plurality of inputs from the plurality of sensors **40** including x, y, and z axis acceleration and angular rate information from accelerometer **30** and gyroscope **31**, respectively. Additional exemplary sensors include wheel acceleration and displacement sensors. Further, terrain sensing logic **912** receives an indication of a steering angle of the steering device **226**, an indication of a steering torque, and an indication of a steering velocity from steering system **104**. Additionally, terrain sensing logic **912** receives an indication of an ambient weather event. In the illustrated embodiment, the ambient weather event is an ambient air temperature and the indication is received from a temperature sensor **194**. The temperature sensor **194** is supported by vehicle **10**. Alternatively, the ambient air temperature may be received from a sensor remote from vehicle **10** and communicated to vehicle **10** over one or more wireless networks, such as over a network from a smartphone, a satellite network, or a cellular network. Terrain sensing logic may further receive engine speed, accelerator pedal position, and vehicle speed values from one or more of the controllers of vehicle **10**. By sensing terrain changes, an operator of vehicle **10** will feel more comfortable driving vehicle **10** in changing terrains without needing to make a ride mode change selection.

[0174] In one embodiment, terrain sensing logic **912** determines vehicle **10** is traversing a harsh trail terrain (chatter) by performing a frequency analysis of incoming inertial measurement information and weighing magnitudes of acceleration at certain frequencies against a calibrated profile. In response thereto, terrain sensing logic **912**, adjusts adjustable shocks **18** to optimize either ride or handling based on the weighted observations.

[0175] In one embodiment, terrain sensing logic **912** determines vehicle **10** is traversing a terrain with small hills (whoops) by measuring repeated bottom out events. In response thereto, terrain sensing logic **912**, adjusts adjustable shocks **18** to increase compression damping.

[0176] In one embodiment, terrain sensing logic **912** determines vehicle **10** is traversing a terrain with ice by measuring high levels of yaw rate in absence of lateral accelerations. In response thereto, terrain sensing logic **912**, adjusts adjustable shocks **18** to increase the damping characteristics, either compression, rebound, or compression and rebound, over current levels.

[0177] In one embodiment, terrain sensing logic **912** determines vehicle **10** is traversing a terrain which causes the vehicle to bottom out by receiving repeated high g-force events in the z-axis (see FIG. **2**) from accelerometer **30**. In response thereto, terrain sensing logic **912**, adjusts adjustable shocks **18** to increase the compression damping of each of the adjustable shocks **18** to reduce the G forces in the z-axis. As the terrain smooths out (i.e. lower G forces in the z-axis), terrain sensing logic **912** decreases the compression damping of each of the adjustable shocks **18** back to a level, such as the current ride mode level.

[0178] In one embodiment, terrain sensing logic **912** adjusts the damping characteristics of adjustable shocks **18** over time based on sensed terrain changes. Referring to FIGS. **27A** and **27B**, in one example, vehicle **200** is traversing a terrain **940** including a trail terrain **950** followed by a road terrain **952**. Exemplary trail terrains include dirt, off-road trails having uneven surfaces. Exemplary road terrains include asphalt surfaces having a generally smooth surface. In the illustrated example, trail terrain **950** includes a whoops portion **954** which includes a plurality of hills that vehicle **200** must traverse.

[0179] Referring to FIG. **27B**, a suspension damping level **960** of at least one of the adjustable shocks **18** is shown over time. At the beginning of the illustrated time period, the suspension damping level remains generally constant as vehicle **200** traverses the trail terrain. Of course, the

damping level may be altered by controller **20** if vehicle **200** is cornering, braking, accelerating, or experiencing other conditions. When vehicle **200** enters the whoops region, the terrain sensing logic **912** of controller **20** increases the damping level of the adjustable shock **18**. In one example, terrain sensing logic **912** increases the damping level due to an increase in the adjustable shock **18** bottoming out. Once **200** exits the whoops portion, terrain sensing logic **912** of controller **20** begins to reduce the damping level. In one example, terrain sensing logic **912** reduces the damping level due to a decrease in the vertical acceleration measured by accelerometer **30**. As shown in FIG. 27B, terrain sensing logic **912** further reduces the damping level when vehicle **200** is traversing road terrain **952**.

[0180] In one embodiment, terrain sensing logic **912** stores the suspension damping levels for later recall to proactively adjust the damping level of adjustable shocks **18**. As discussed herein, vehicle **200** may include a location determiner **110**. In one example, terrain suspension logic **912** stores the suspension damping levels based on vehicle location as determined by location determiner **110**. Thus, when vehicle **200** is approaching the stored location, terrain sensing logic **912** will adjust the suspension damping level based on the stored suspension damping level. In a further example, terrain suspension logic **912** will take into account the current vehicle speed when adjusting the suspension damping levels.

[0181] While embodiments of the present disclosure have been described as having exemplary designs, the present invention may be further modified within the spirit and scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the disclosure using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this invention pertains.

Claims

1. A recreational vehicle, comprising: a plurality of ground engaging members; a frame supported by the plurality of ground engaging members; at least one suspension coupling the plurality of ground engaging members to the frame, the at least one suspension including at least one adjustable shock absorber having at least one adjustable damping profile; a driveline torque management system operatively coupled to at least one of the plurality of ground engaging members, the driveline torque management system having at least one adjustable driveline torque management system characteristic; at least one airborne sensor positioned on the recreational vehicle, the at least one airborne sensor monitoring a movement characteristic of the recreational vehicle; and at least one controller operatively coupled to the driveline torque management system, the at least one airborne sensor, and the at least one adjustable shock absorber, the at least one controller determining if the recreational vehicle is in an airborne state and in response to determining the recreational vehicle is in the airborne state altering both the at least one adjustable damping profile of the at least one adjustable shock absorber and altering the at least one adjustable driveline torque management system characteristic of the driveline torque management system.
2. The recreational vehicle of claim 1, wherein the at least one airborne sensor is supported by the frame and spaced apart from the at least one adjustable shock absorber.
3. The recreational vehicle of claim 2, wherein the at least one airborne sensor is positioned generally along a longitudinal centerline plane of the recreational vehicle.
4. The recreational vehicle of claim 3, wherein the at least one airborne sensor includes an accelerometer.
5. The recreational vehicle of claim 4, wherein the at least one airborne sensor includes a gyroscope.
6. The recreational vehicle of claim 1, wherein the at least one adjustable driveline torque management system characteristic of the driveline torque management system is associated with a

prime mover of the driveline torque management system.

7. The recreational vehicle of claim 6, wherein the at least one adjustable driveline torque management system characteristic associated with the prime mover of the driveline torque management system is adjusted to alter a pitch of the recreational vehicle.

8. The recreational vehicle of claim 1, wherein the at least one adjustable driveline torque management system characteristic of the driveline torque management system is associated with a transmission of the driveline torque management system.

9. The recreational vehicle of claim 8, wherein the at least one adjustable driveline torque management system characteristic associated with the transmission of the driveline torque management system is adjusted to alter a pitch of the recreational vehicle.

10. The recreational vehicle of claim 1, wherein the at least one adjustable driveline torque management system characteristic of the driveline torque management system is associated with a braking system of the driveline torque management system.

11. The recreational vehicle of claim 10, wherein the at least one adjustable driveline torque management system characteristic associated with the braking system of the driveline torque management system is adjusted to alter a pitch of the recreational vehicle.

12. The recreational vehicle of claim 1, wherein the at least one airborne sensor includes an accelerometer which provides an x-axis acceleration value, a y-axis acceleration value, and a z-axis acceleration value, the at least one controller determining if the recreational vehicle is in an airborne state by comparing a magnitude of a resultant acceleration vector to a first threshold, the resultant acceleration vector being based on the x-axis acceleration value, the y-axis acceleration value, and the z-axis acceleration value.

13. The recreational vehicle of claim 1, wherein the at least one airborne sensor includes an accelerometer which provides an x-axis acceleration value, a y-axis acceleration value, and a z-axis acceleration value, the at least one controller determining if the recreational vehicle is in an airborne state by comparing the x-axis acceleration value to a first threshold value, the y-axis acceleration value to a second threshold, and the z-axis acceleration value to a third threshold.

14. A recreational vehicle, comprising: a plurality of ground engaging members; a frame supported by the plurality of ground engaging members; at least one suspension coupling the plurality of ground engaging members to the frame, the at least one suspension including at least one adjustable shock absorber having at least one adjustable damping profile; a driveline torque management system operatively coupled to at least one of the plurality of ground engaging members, the driveline torque management system having at least one adjustable driveline torque management system characteristic; at least one airborne sensor positioned on the recreational vehicle, the at least one airborne sensor monitoring a movement characteristic of the recreational vehicle; and at least one controller operatively coupled to the driveline torque management system, the at least one airborne sensor, and the at least one adjustable shock absorber, the at least one controller determining if the recreational vehicle is in an airborne state and in response to determining the recreational vehicle is in the airborne state altering the at least one adjustable damping profile of the at least one adjustable shock absorber to a first damping profile if the airborne state spans a first time duration and to a second damping profile if the airborne state spans a second time interval, the second time interval being longer than the first time duration.

15. The recreational vehicle of claim 14, wherein the second damping profile is stiffer than the first damping profile.

16. The recreational vehicle of claim 14, wherein the at least one adjustable driveline torque management system characteristic associated with a transmission of the driveline torque management system is adjusted to alter a pitch of the recreational vehicle.

17. The recreational vehicle of claim 14, wherein the at least one airborne sensor includes an accelerometer.

18. The recreational vehicle of claim 14, wherein the at least one controller is configured to receive

an output from one or more sensors associated with a magnitude of acceleration along an axis.

19. The recreational vehicle of claim 14, wherein the at least one controller is configured to change a pitch of the recreational vehicle through the driveline torque management system.

20. The recreational vehicle of claim 14, wherein the at least one controller is configured to record information related to information from the at least one airborne sensor.
